

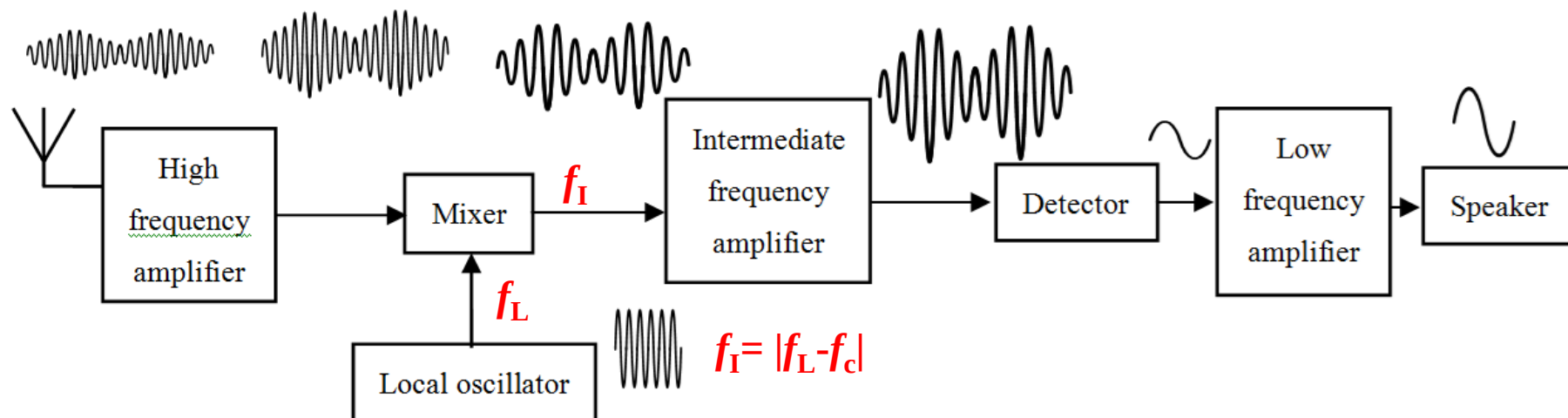


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# Communication electronics circuits

## Chapter 1

# Super heterodyne receiving system



**AM radio:  $f_I = 465$  kHz;**

**FM radio:  $f_I = 10.7$  MHz;**

**TV:  $f_I = 38$  MHz;**

**microwave receiver and satellite receiver:  $f_I = 70$  MHz or 140 MHz**

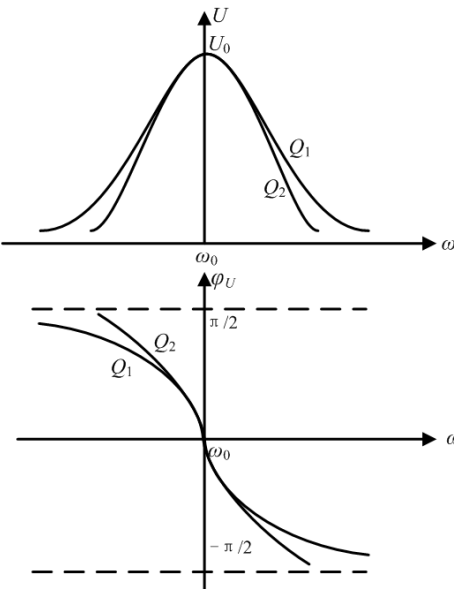
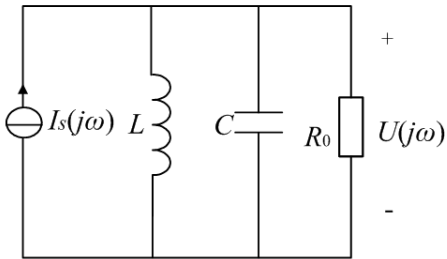
# Divisions of radio frequency (wave) bands

| Band            | Frequency range | Wavelength  | Frequency   |
|-----------------|-----------------|-------------|-------------|
| Very-long wave  | VLF             | 100~10km    | 3~30KHz     |
| Long wave       | LF              | 10~1km      | 30~300KHz   |
| Medium wave     | MF              | 1000~100m   | 300KHz~3MHz |
| Short wave      | HF              | 100~10m     | 3~30MHz     |
| Very-short wave | VHF             | 10~1m       | 30~300MHz   |
| Decimeter wave  | UHF             | 1~0.1m      | 300~3000MHz |
| Centimeter wave | SHF             | 10~1cm      | 3~30GHz     |
| Millimeter wave | EHF             | 10~1mm      | 30~300GHz   |
| Light wave      |                 | 300~0.006μm | 1~50THz     |

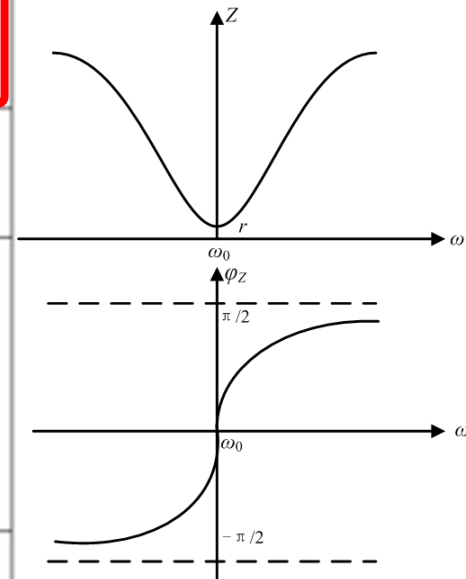
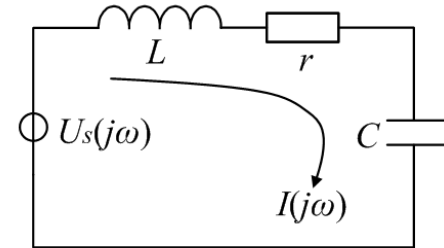
$$\lambda = \frac{c}{f}$$



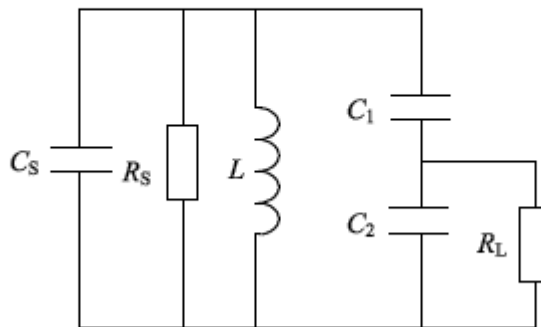
# Comparison of LC parallel, series circuits



|          | 并联谐振回路                                                                                                                                                     | 串联谐振回路                                                                                                                                                     |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 谐振角频率    | $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$                                                                                                                  | $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$                                                                                                                  |
| 品质因数     | $Q_0 = \frac{R_0}{\omega_0 L} = R_0 \omega_0 C$                                                                                                            | $Q_0 = \frac{\omega_0 L}{r} = \frac{1}{r \omega_0 C}$                                                                                                      |
| 阻抗       | 谐振时最大, 为纯阻性                                                                                                                                                | 谐振时最小, 为纯阻性                                                                                                                                                |
| 谐振电阻     | $R_0 = Q_0 \omega_0 L = \frac{Q_0}{\omega_0 C} = \frac{L}{Cr}$                                                                                             | $r = \frac{1}{Q_0} \sqrt{\frac{L}{C}}$                                                                                                                     |
| 失谐时阻抗特性  | $f > f_0$ 容性<br>$f < f_0$ 感性                                                                                                                               | $f > f_0$ 感性<br>$f < f_0$ 容性                                                                                                                               |
| 通频带      | $B_{0.7} = \frac{f_0}{Q}$                                                                                                                                  | $B_{0.7} = \frac{f_0}{Q}$                                                                                                                                  |
| 归一化选频特性  | $ \alpha  = \frac{1}{\sqrt{1 + Q^2 \left( \frac{f}{f_0} - \frac{f_0}{f} \right)^2}}$<br>$\varphi = -\arctg Q \left( \frac{f}{f_0} - \frac{f_0}{f} \right)$ | $ \alpha  = \frac{1}{\sqrt{1 + Q^2 \left( \frac{f}{f_0} - \frac{f_0}{f} \right)^2}}$<br>$\varphi = -\arctg Q \left( \frac{f}{f_0} - \frac{f_0}{f} \right)$ |
| 负载和内阻的影响 | $Q \downarrow$                                                                                                                                             | $Q \downarrow$                                                                                                                                             |



In Fig.2,  $L = 0.8 \mu\text{H}$ ,  $C_1 = 20 \text{ pF}$ ,  $C_S = 10 \text{ pF}$ ,  $R_S = 20 \text{ k}\Omega$ ,  $R_L = 5 \text{ k}\Omega$ ,  $Q_0 = 100$ . If  $R_L$  and  $R_S$  achieves impedance matching at resonant frequency, what the value of  $C_2$ ? Calculate the resonant frequency  $f_0$ , resonant resistance  $R_0$ ,  $Q_L$  and  $B_{0.7}$ .



**Fig. 2**

$$n = \frac{C_1}{C_1 + C_2}$$

$$R'_L = \frac{R_L}{n^2}$$

Since  $R_L$  and  $R_S$  achieves impedance matching, so

$$R'_L = R_S \cdot n = 0.5$$

$$C_2 = 20 \text{ pF}$$

$$C_\Sigma = C_S + \frac{C_1 \times C_2}{C_1 + C_2} = 20 \text{ pF}$$

$$\omega_0 = \frac{1}{\sqrt{LC_\Sigma}} = 0.25 \times 10^9 \text{ rad / s}$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 39.8 \text{ MHz}$$

Since  $Q_0$  is finite; so the inductor has a parasitic shunted resistance,

$$R_0 = Q_0 \omega_0 L = 20 \text{ k}\Omega$$

$$R_\Sigma = R_S \parallel R'_L \parallel R_0 = 6.67 \text{ k}\Omega$$

$$Q_L = \omega_0 R_\Sigma C_\Sigma = 33.3$$

$$B_{0.7} = \frac{f_0}{Q_L} = 1.2 \text{ MHz}$$



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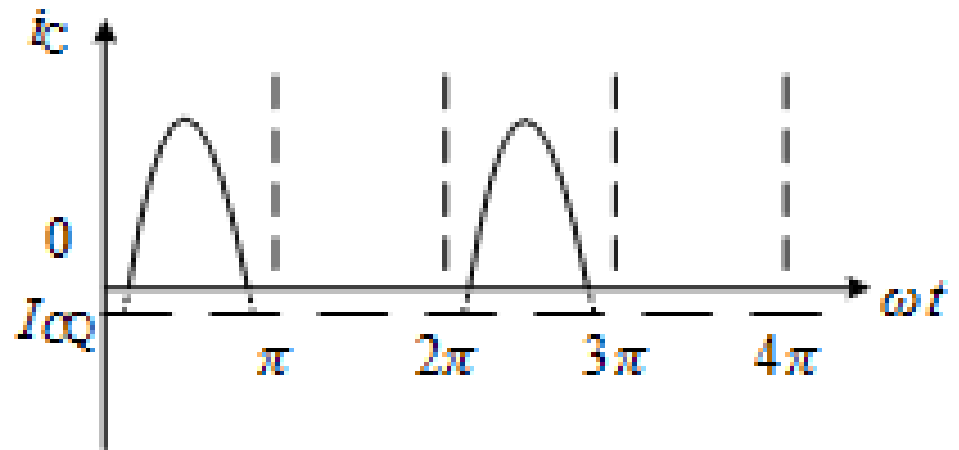
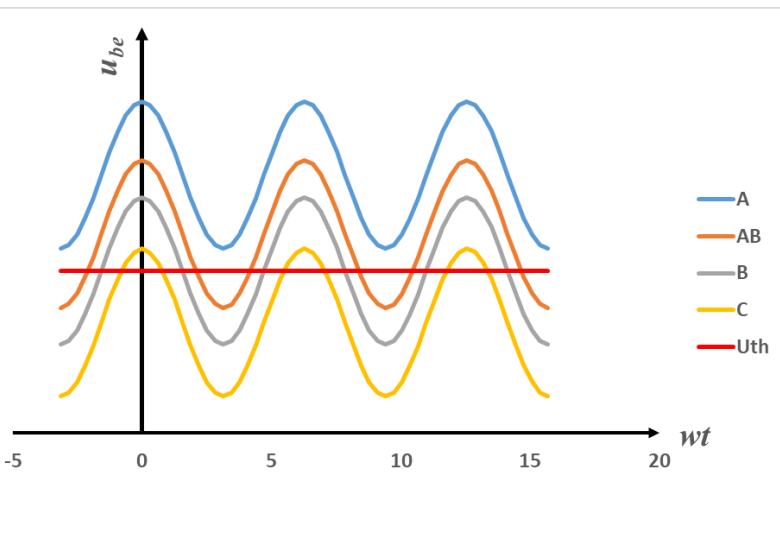
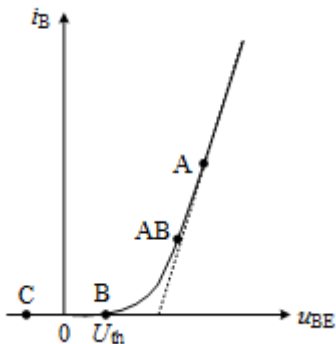
Communication electronics circuits

## Chapter 2

# Radio Frequency Amplifier

# Classification of Power Amplifier

conduction angle  $2\theta$



(d) Class C

$$2\theta < 180^\circ$$

# Performance of RFPA

## ◆ Output Power

$$P_{out} = \frac{1}{2} U_m I_m = \frac{1}{2} I_m^2 R_c = \frac{1}{2} \frac{U_m^2}{R_c}$$

$$P_{in} = \frac{1}{2\pi} \int_0^{2\pi} i_B u_{BE} d(\omega t)$$

## ◆ Power Gain

$$G_p = \frac{P_{out}}{P_{in}}$$

*broad dynamic range signal*

$$P_W = 10^{\frac{P_{dBm}}{10} - 3} W$$

$$G_{dB} = 10 \cdot \log_{10}(G_{dB}) = P_{out} (dBm) - P_{in} (dBm)$$

## ◆ Efficiency

$$\eta_c = \frac{P_{out}}{P_{DC}} = \frac{\frac{1}{2} U_m I_m}{E_c I_0} = \frac{1}{2} \frac{U_m}{E_c} \frac{I_m}{I_0} = \frac{1}{2} \xi \gamma$$

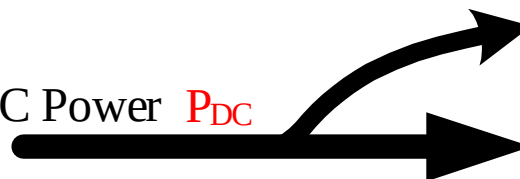
$$\xi = U_m / E_c \quad \gamma = I_m / I_0$$

$P_{DC}$ : the source DC power

DC Power  $P_{DC}$

Transistor Dissipate  
Power  $P_C$

Output Power  $P_{out}$





# Performance of RFPA

- ◆ The net increase in the signal power from input to its output:

$$P_{\text{add}} = P_{\text{out}} - P_{\text{in}}$$

- ◆ Power added efficiency (PAE)

$$PAE = \eta_{\text{add}} = \frac{P_{\text{add}}}{P_{\text{DC}}} = \frac{P_{\text{out}} - P_{\text{in}}}{P_{\text{DC}}} = \frac{P_o \cdot (1 - \frac{1}{G_p})}{P_{\text{DC}}} = \eta \cdot (1 - \frac{1}{G_p})$$

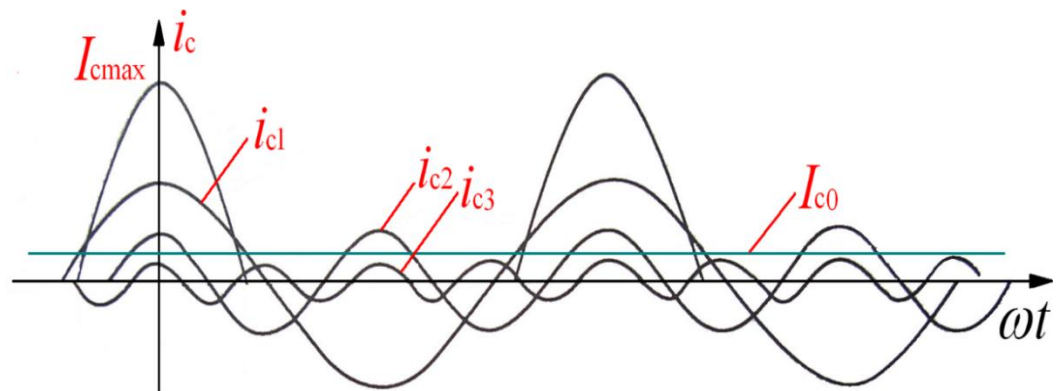
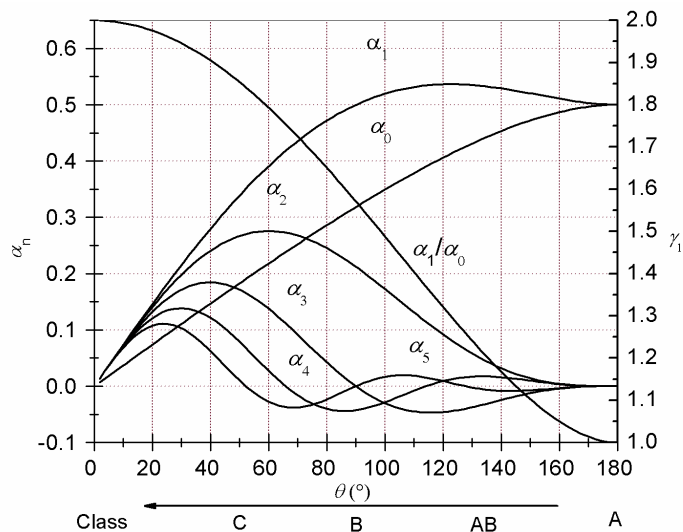
$G_p$  is the operating power gain

# Analysis of periodic cosine pulses

$$i_c(\omega t) = I_{c\max} \left[ \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos n\omega t \right] \quad \left\{ \begin{array}{l} I_{c0} = \alpha_0 I_{c\max} \rightarrow \text{DC} \rightarrow P_{DC} = E_C \cdot I_{c0} \\ I_{c1} = \alpha_1 I_{c\max} \rightarrow \text{Fundamental} \rightarrow P_{out} = \frac{1}{2} I_{c1} U_{c1m} \end{array} \right.$$

⋮

$$u_c(\omega t) = U_{c1m} + E_C \quad \left\{ \begin{array}{l} E_C \rightarrow \text{DC} \\ U_{c1m} \rightarrow \text{Fundamental} \rightarrow \max U_{c1m} = E_C - U_{ces} \end{array} \right.$$



# Calculation of Collection Efficiency $\eta_c$

◆ Voltage source providing power  $P_{DC} = E_C \cdot I_{c0}$

◆ Output power of transistor collection

$$P_{out} = \frac{1}{2} U_{c1m} I_{c1m} = \frac{1}{2} \frac{U_{c1m}^2}{R_c} = \frac{1}{2} I_{c1m}^2 R_c$$

$$R_c = \frac{U_{c1m}^2}{2 P_{out}}$$

◆ Output efficiency of collector

$$\begin{aligned} \eta_c &= \frac{P_{out}}{P_{DC}} = \frac{1}{2} \frac{U_{c1m} \cdot I_{c1m}}{E_C \cdot I_{c0}} = \frac{1}{2} \frac{U_{c1m} I_{c \max} \alpha_1(\theta)}{E_C I_{c \max} \alpha_0(\theta)} \\ &= \frac{1}{2} \frac{U_{c1m}}{E_C} \frac{\alpha_1(\theta)}{\alpha_0(\theta)} = \frac{1}{2} \xi \gamma(\theta) \end{aligned}$$

$$\xi = \frac{U_{c1m}}{E_C} \quad \text{Output (collector) voltage factor}$$

$$\gamma(\theta) = \frac{\alpha_1(\theta)}{\alpha_0(\theta)} \quad \text{Output (collector) current factor}$$

◆ Dissipated power of transistor

$$P_C = P_{DC} - P_{out} = \frac{P_{out}}{\eta_c} - P_{out}$$

# 3 Kinds of Operating Status of PA

$$\blacksquare u_{\text{cemin}} = U_{\text{ces}}$$

Critical state (CS)

optimized status!

$$\blacksquare u_{\text{cemin}} < U_{\text{ces}}$$

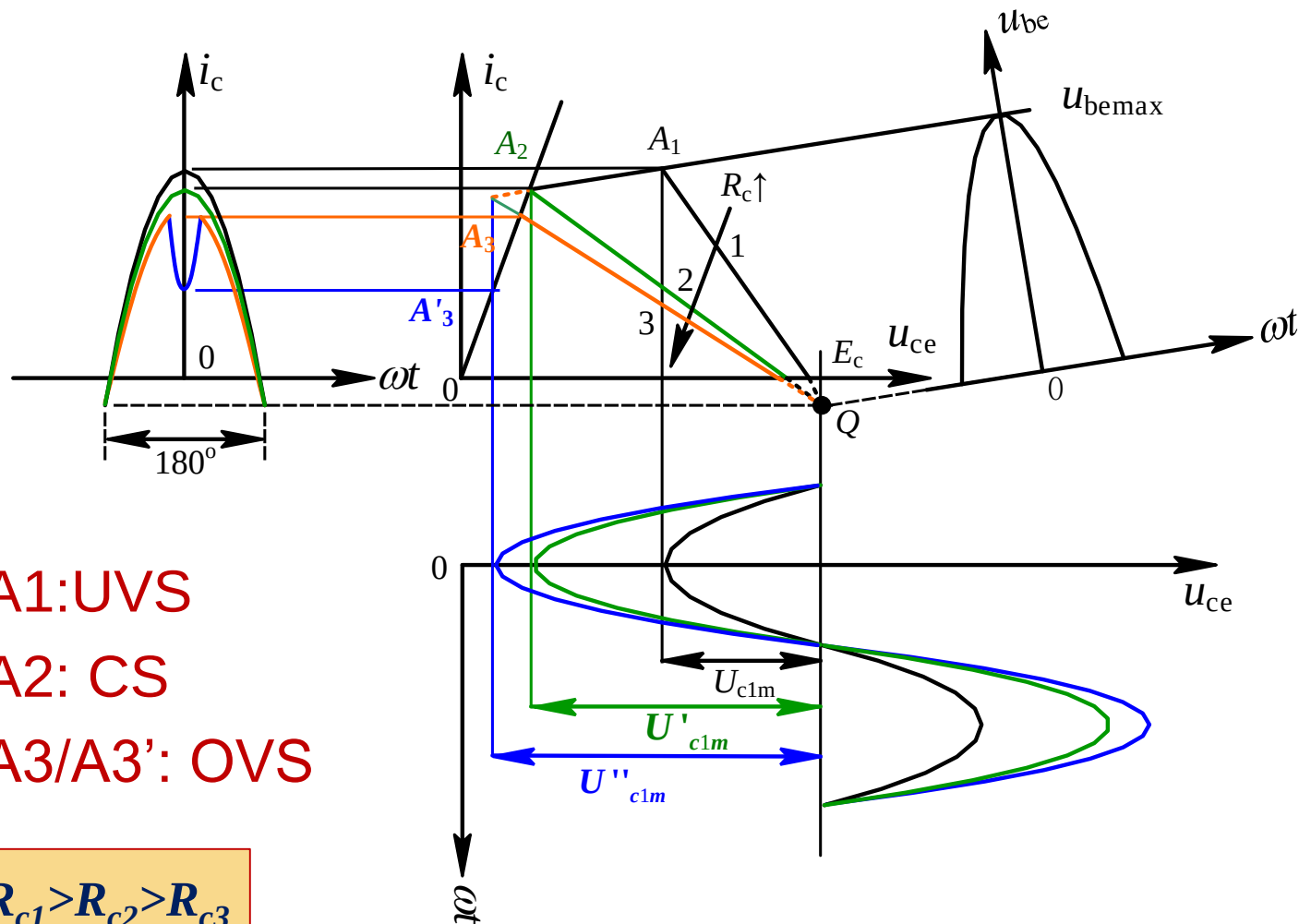
Over voltage state (OVS)

$$\blacksquare u_{\text{cemin}} > U_{\text{ces}}$$

Under voltage state (UVS)

- ◆ **Under-Voltage Status:** the efficiency  $\eta_c$  is low, the transistor dissipate  $P_C$  is high, the transistor is risk of burnout,  $P_{CM} > P_C$ .
- ◆ **Over-voltage Status:** In the low over-voltage status, the efficiency reach maximum  $\eta_{\text{cmax}}$ , deep over-voltage status, waveform  $i_c$  is sunken.

# Rc Changes: Load Characteristic



A1: UVS

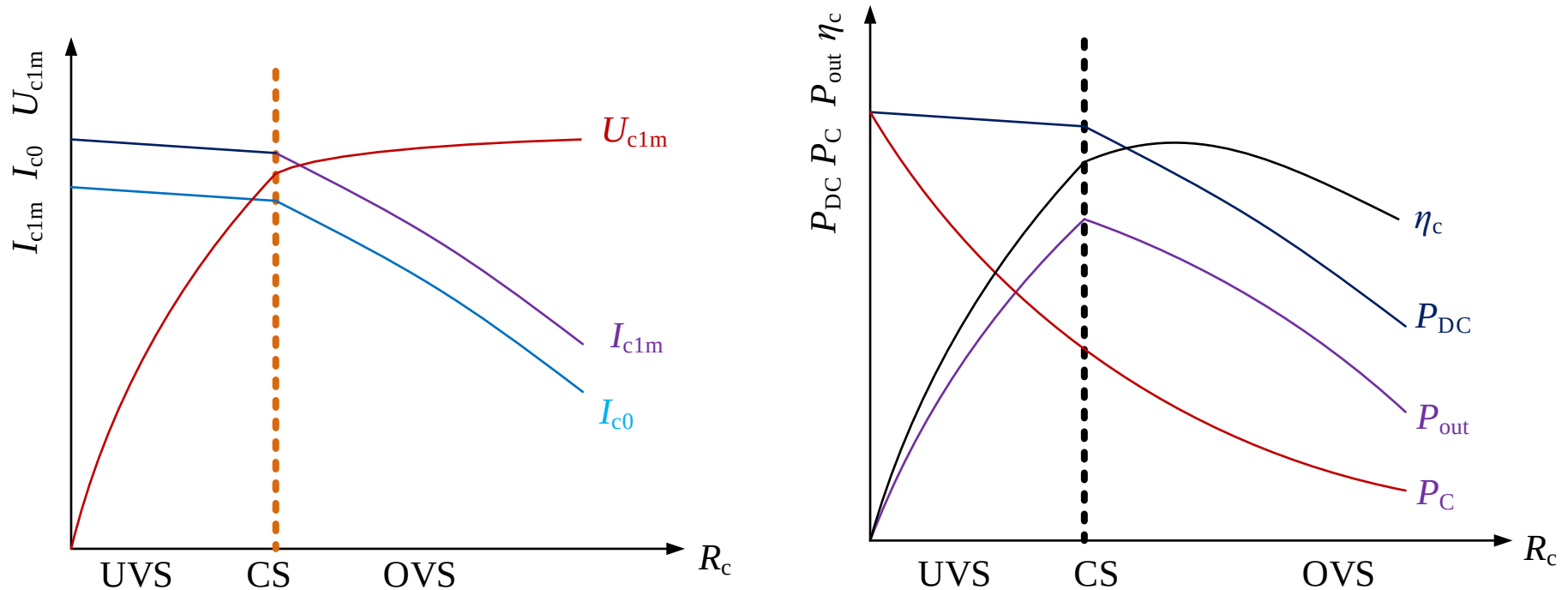
A2: CS

A3/A3': OVS

$$R_{c1} > R_{c2} > R_{c3}$$

(1)  $\eta_c$  is low,  $P_C$  is high, the transistor burnout,  $P_{CM} > P_C$ . The load must not be shorted.

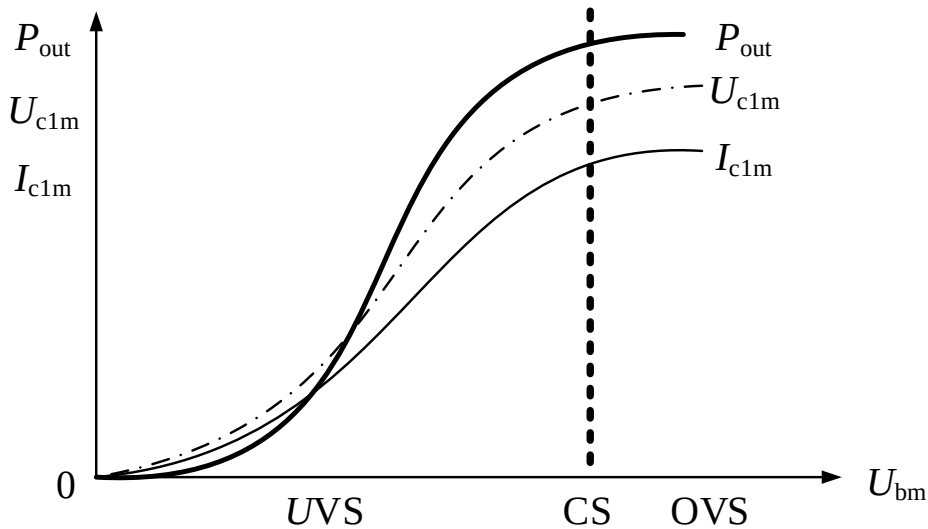
# Load characteristic of tuned PA



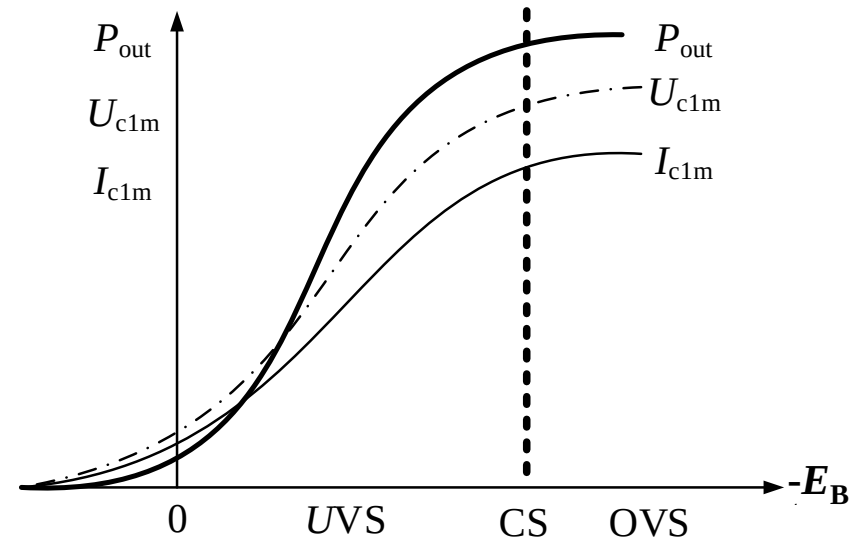
- ◆ **Under-Voltage Status:**  $\eta_c$  is low,  $P_C$  is high, the transistor **burnout**,  $P_{CM} > P_C$ . **The load must not be shorted.**
- ◆ **Critical Point:**  $P_{out}$  is maximum,  $\eta_c$  is high, **optimized status!**
- ◆ **Over-voltage Status:** low over-voltage status, the efficiency reach maximum  $\eta_{cmax}$ .

# $U_{bm}$ Changes: Amplitude Characteristic

$$u_{bemax} = -E_B + U_{bm}$$



Amplitude characteristic

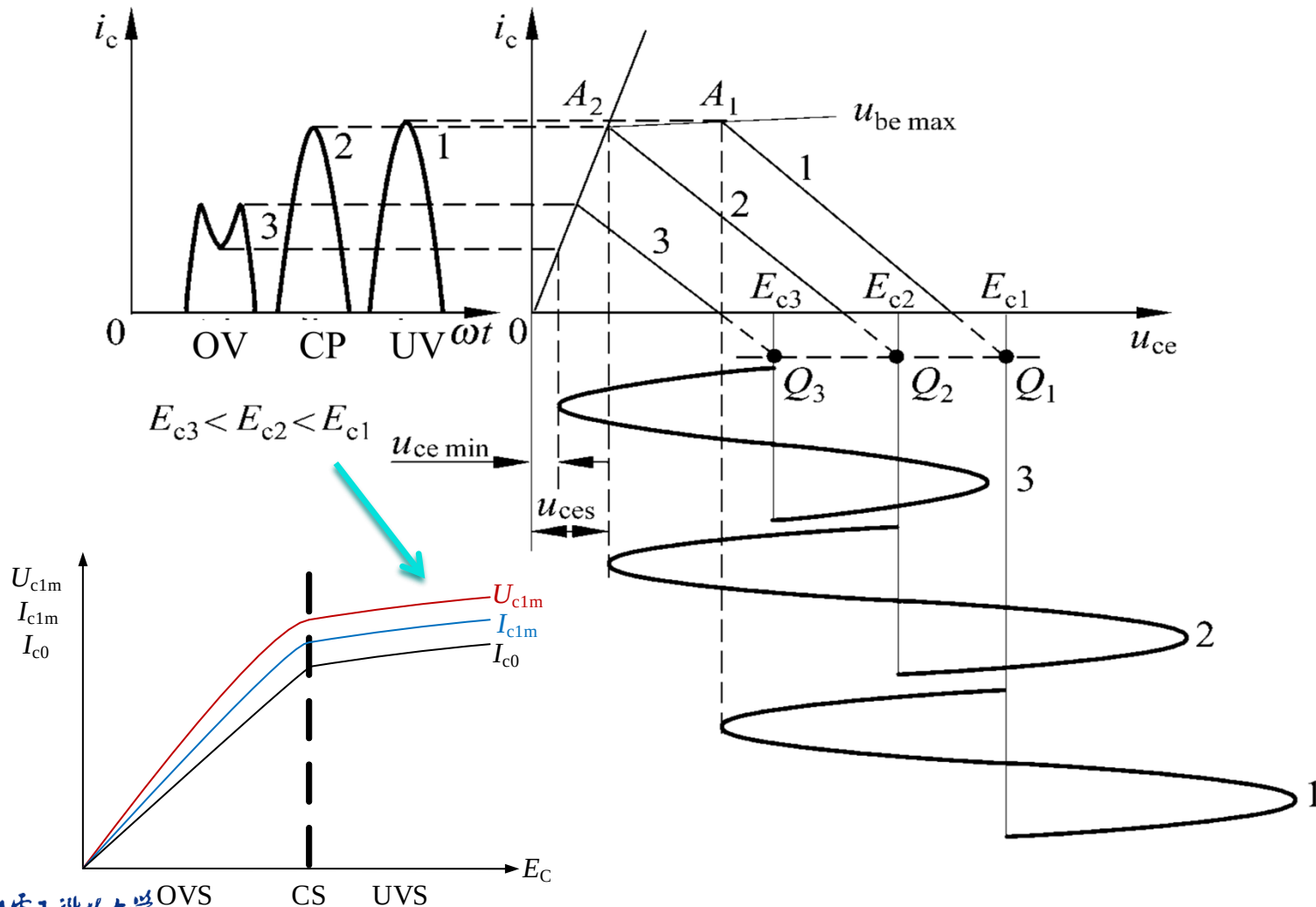


Base modulation characteristic

# Ec Changes: Collector Modulation Characteristic

$$u_{ce\min} = E_C - U_{c1m} = E_C - I_{c1m}R_C = E_C - I_{c\max}\alpha_1(\theta)R_C$$

$$u_{be\max} = -E_B + U_{bm}$$





A resonant power amplifier with operating at critical point state. Assuming  $E_c = 24V$ , current conduction angle (CCA)  $\theta = 70^\circ$ ,  $\alpha_0(70^\circ) = 0.253$ ,  $\alpha_1(70^\circ) = 0.436$ , collector current pulse maximum value  $I_{c\max} = 2.2A$ , collector voltage utilization factor  $\xi=0.9$ . Please calculate output power  $P_o$ , voltage source providing power  $P_s$ , collector efficiency  $\eta_c$  and collector equivalent load resistor  $R_c$ .

$$U_{c1m} = \xi \times E_c = 0.9 \times 24 = 21.6V$$

$$\eta_c = \frac{P_o}{P_s} = \frac{10.36}{13.36} \approx 77.5\%$$

$$I_{c1m} = \alpha_1(\theta) \times I_{c\max} = 0.436 \times 2.2 \approx 0.959A$$

$$P_o = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 21.6 \times 0.959 \approx 10.36W$$

$$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{21.6}{0.959} \approx 22.5\Omega$$

$$I_{c0} = \alpha_0(\theta) \times I_{c\max} = 0.253 \times 2.2 = 0.5566A$$

$$P_s = E_c I_{c0} = 24 \times 0.5566 \approx 13.36W$$



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Communication electronics circuits

# Chapter 3

## Sinusoidal Oscillator

# Principle of feedback LC oscillator

## 1. Feedback *LC* oscillator:

three ends of ***LC* circuit** are separately connected with three electrodes of **transistor**.

## 2. Rule of Composition:

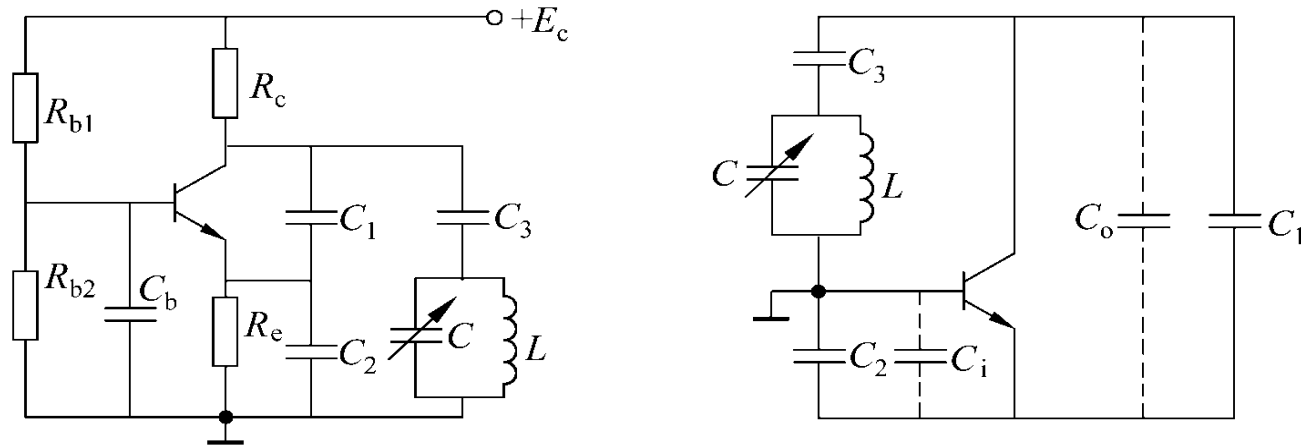
**Reactance components connected to the emitter are the same and contrary to the one between the collector and the base.**

**(射同基集反)**

## 3. Three kinds of feedback LC oscillator :

- ❑ Capacitor Feedback LC Oscillator (Colpitts Circuit)
- ❑ The First Improved Capacitor feedback *LC* oscillator (Clapp Circuit)
- ❑ The Second Improved Capacitor Feedback LC Oscillator ( Seiler Circuit)
- ❑ Inductor Feedback LC Oscillator (Hartley Circuit )

# The Second Improved Capacitor Feedback LC Oscillator ( Seiler Circuit)

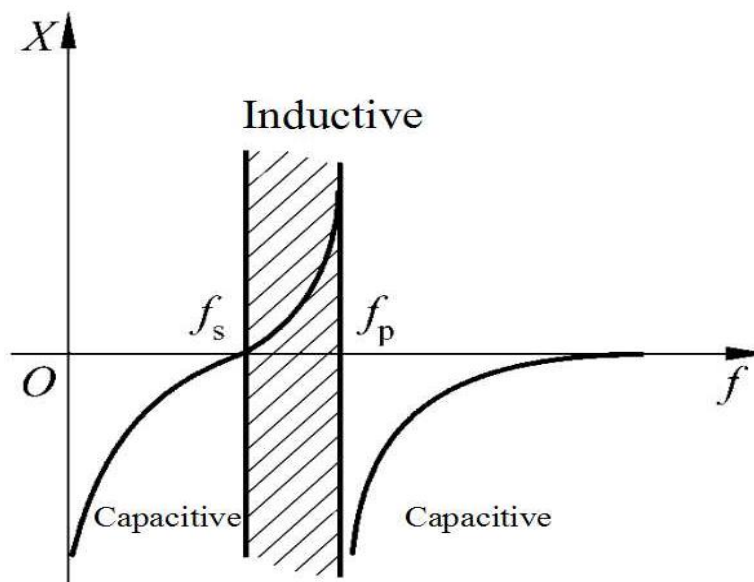
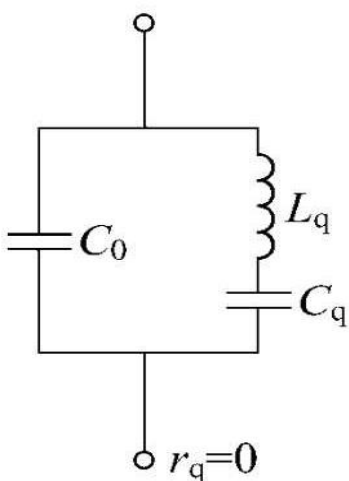


$$\omega_0 \approx \frac{1}{2\pi\sqrt{LC_\Sigma}} \quad C_\Sigma = C + \frac{1}{\frac{1}{C_1 + C_0} + \frac{1}{C_2 + C_i} + \frac{1}{C_3}} \approx C + C_3 \quad F \approx \frac{C_1}{C_1 + C_2}$$

- (1) **Output amplitude is relatively stable;**
- (2) **Tuning range is relatively wide, it has been widely used in broadband system.**

# Quartz Crystal Oscillator

- ◆ **Parallel mode**——The crystal is acting as **an inductor** of the three point type circuit operating at a frequency between  $f_s$  and  $f_p$ .
- ◆ **Series mode**——The crystal is acting as **a short elements** at  $f_s$  in the positive feedback path.



By using the phase equilibrium condition, solve the questions below:

- (1) determine which of the oscillation circuits shown in Fig.2 are possible to oscillate,
- (2) specify the type of oscillation circuits and explain the oscillation conditions.

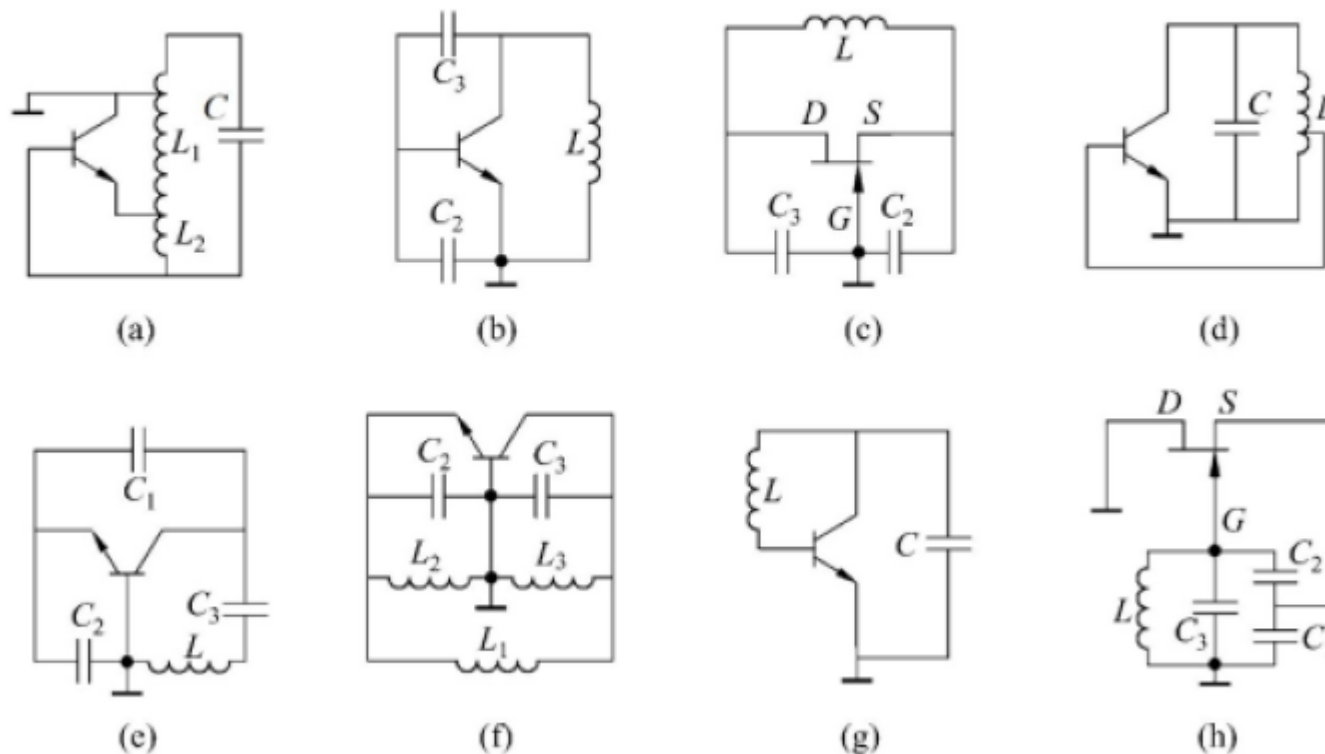
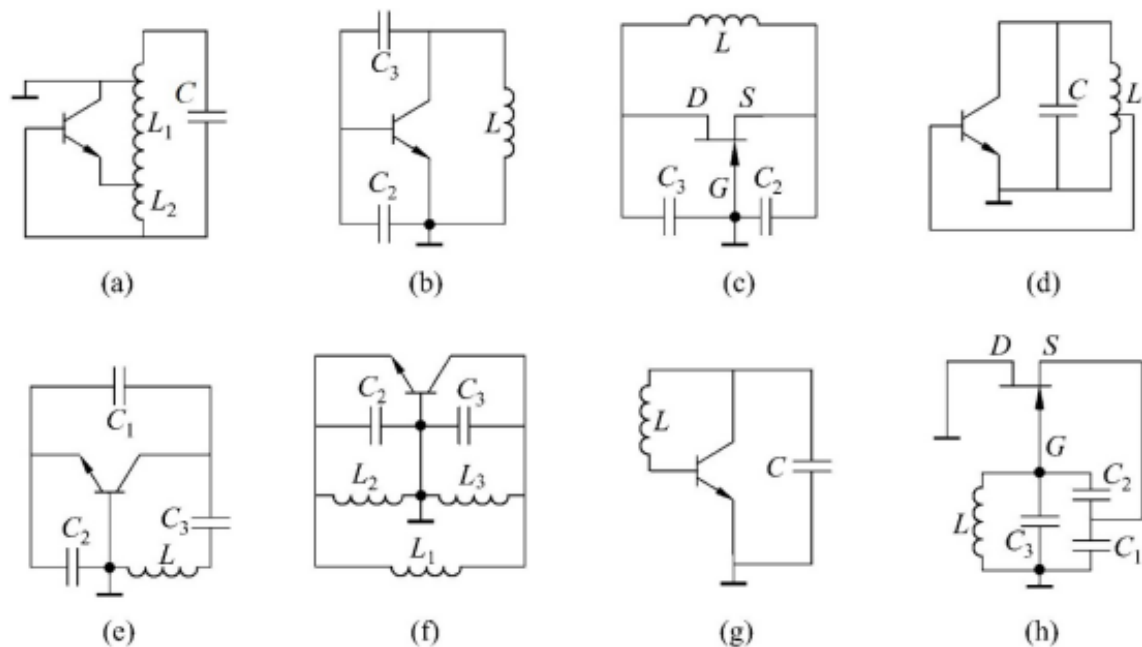


Fig. 2



不能振荡的有：(b) (c) (d)；

电感反馈型振荡器有：(a)、(f)；

其中(f)能够振荡需要  $L_2C_2$  在振荡频率处呈感性， $L_3C_3$  在振荡频率处呈容性，并联谐振低频呈感性，高频呈容性，因此

$$f_{03} = \frac{1}{2\pi\sqrt{L_3C_3}} < f_{osc} < f_{02} = \frac{1}{2\pi\sqrt{L_2C_2}}$$

电容反馈型振荡器有：(e)、(g)、(h)；

其中(e)能够振荡要求  $LC_3$  呈现感性，根据串联谐振低频呈容性，高频呈感性，因此

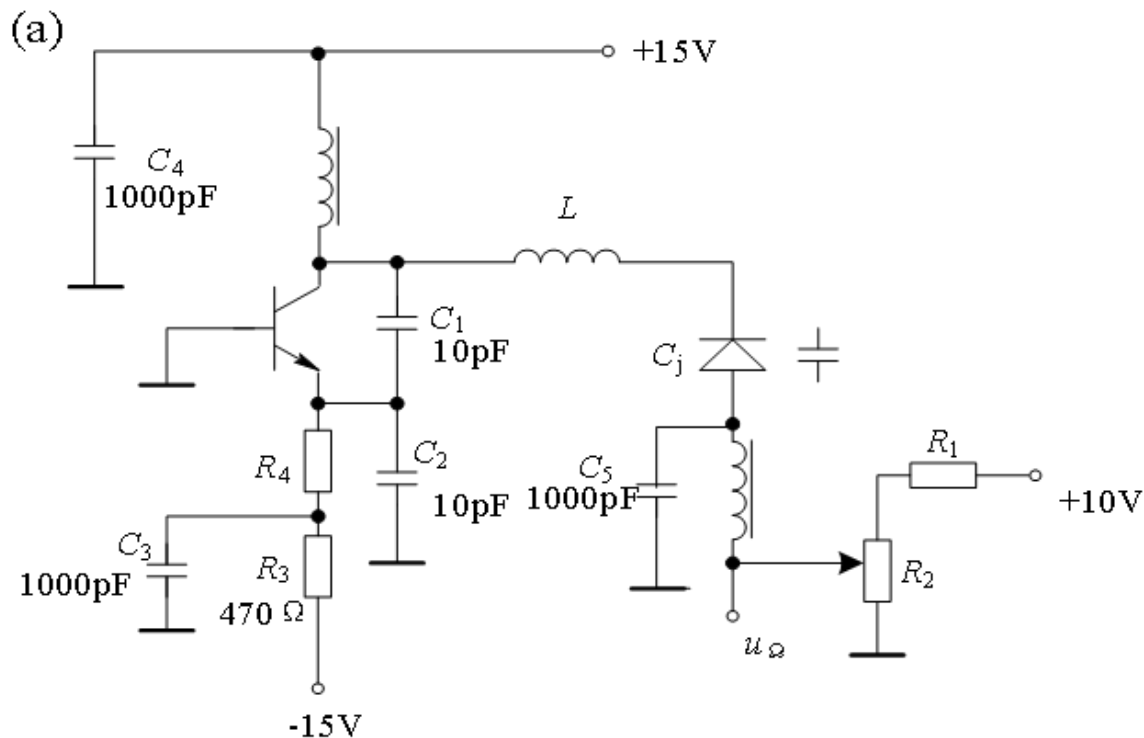
$$f_{osc} > f_{03} = \frac{1}{2\pi\sqrt{LC_3}}$$

其中(g)能够振荡是将基极-发射极间电容作为振荡电容，是密勒振荡器；

其中(h)能够振荡要求  $LC_3$  呈现感性，根据并联谐振高频呈容性，低频呈感性，因此

$$f_{osc} < f_{03} = \frac{1}{2\pi\sqrt{LC_3}}$$

The varactor diode voltage controlled oscillator which center frequency is 100MHz is shown below, please draw the DC channel and high frequency oscillation circuit of the transistor, as well as the varactor's DC bias circuit and low frequency control circuit.







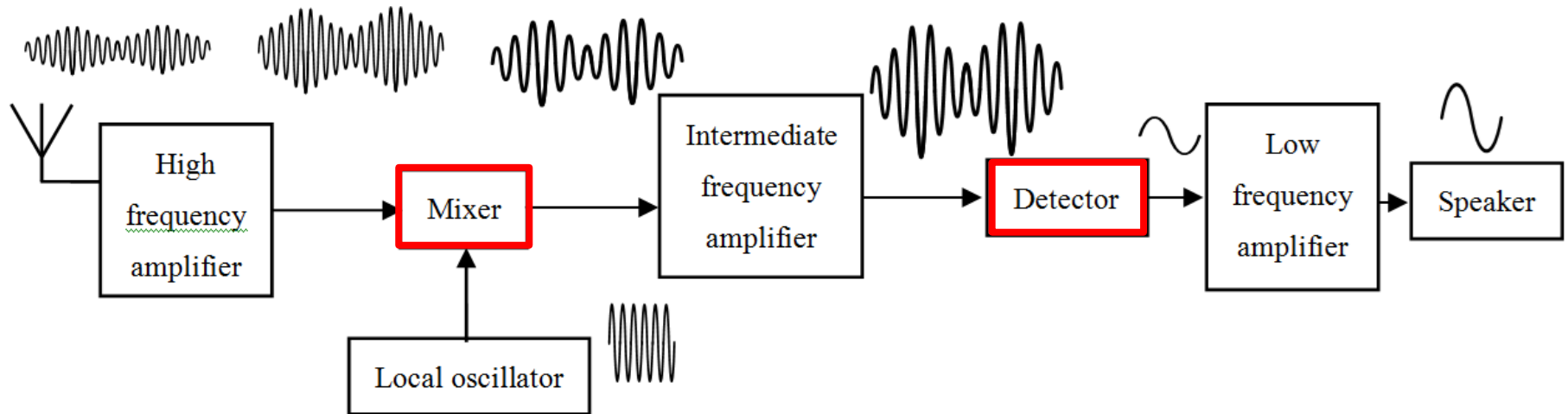
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# Communication electronics circuits

## Chapter 4

# Chapter 4

## Amplitude modulation, Demodulation, Frequency Mixing



### 1. AM theory

- 1) Mathematical expressions
- 2) Frequency spectrum
- 3) Average power

### 2. Amplitude modulated and demodulated circuits

- 1) High level 、 Low level
- 2) Detector
- 3) Mixer

# 1. Standard/DSB/SSB AM signal analysis

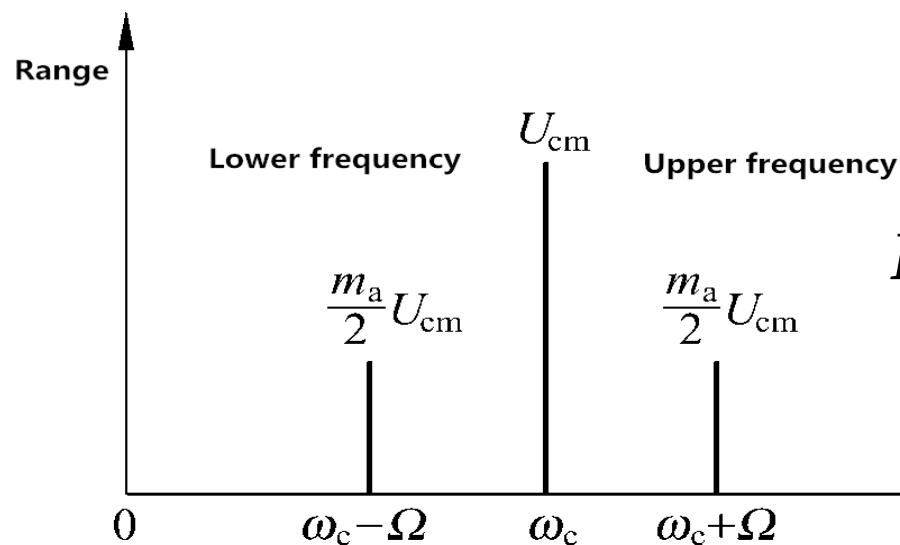


$$u_{\Omega}(t) = U_{\Omega m} \cos \Omega t$$

$$u_c(t) = U_{cm} \cos \omega_c t$$

$$u_{AM}(t) = U_{cm} (1 + m_a \cos \Omega t) \cos \omega_c t$$

$$= U_{cm} \cos \omega_c t + \frac{1}{2} m_a U_{cm} \cos(\omega_c + \Omega)t + \frac{1}{2} m_a U_{cm} \cos(\omega_c - \Omega)t$$

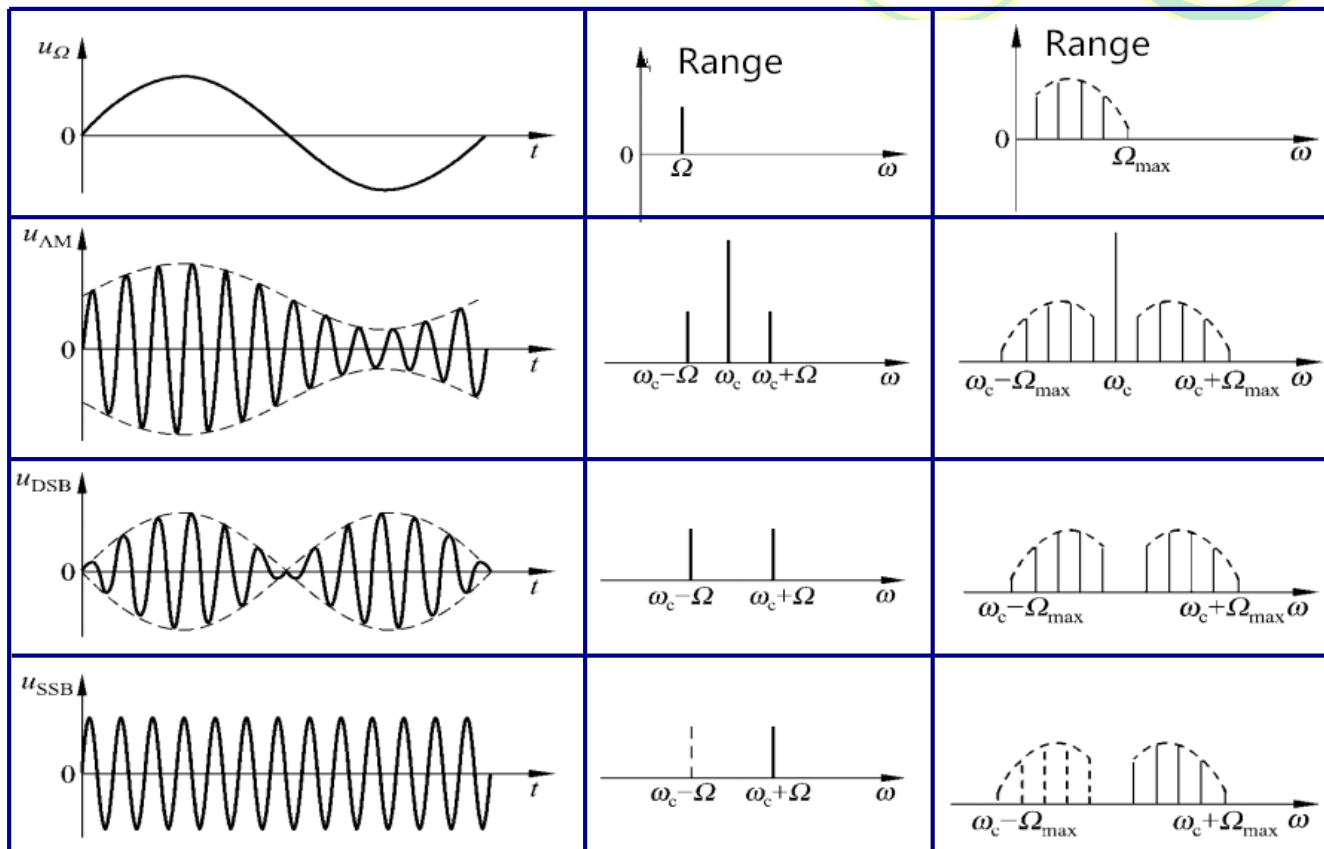


$$P_c = \frac{1}{2} \frac{U_{cm}^2}{R_L}$$

$$P_1 = P_2 = \frac{1}{2} \left( \frac{m_a}{2} U_{cm} \right)^2 \frac{1}{R_L} = \frac{1}{4} m_a^2 P_c$$

$$P = P_c + P_1 + P_2 = \left( 1 + \frac{m_a^2}{2} \right) P_c$$

## 2. AM waveform in time domain, frequency domain



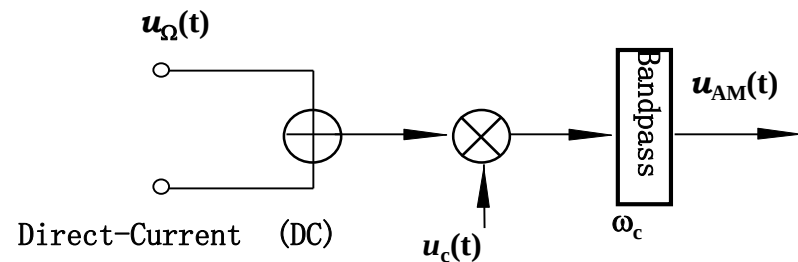
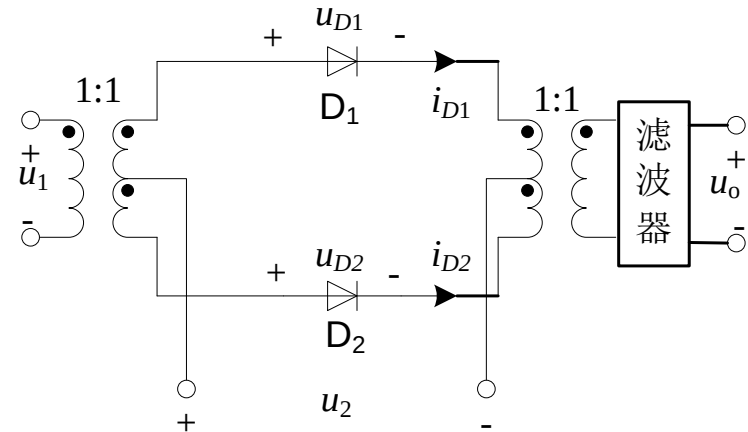
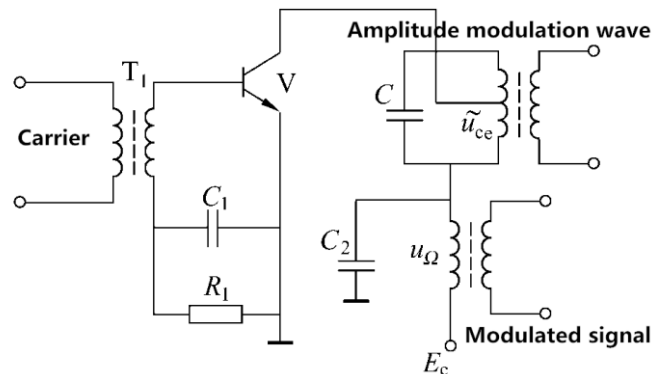
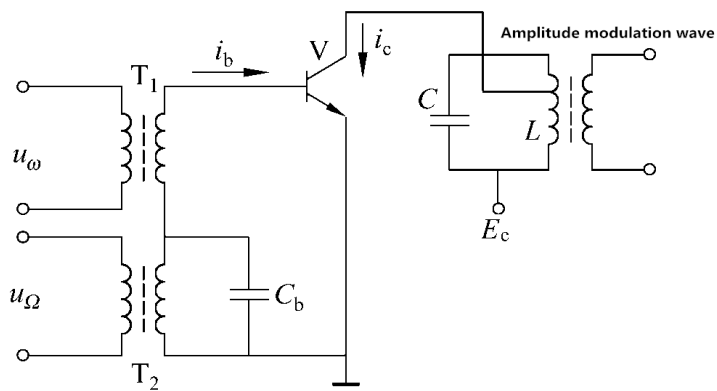
The band-width of multi-frequency modulation AM wave is:

$$B = 2F_{\max}$$

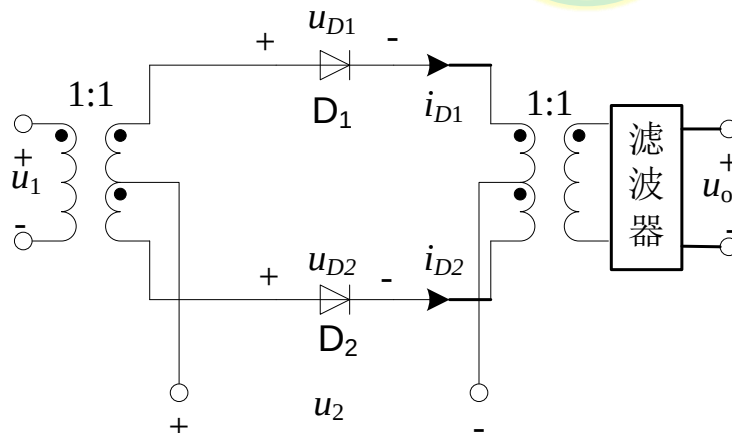
# 3 Amplitude-modulated circuits



- 1) **High level modulation** - final stage high power AM wave;  
Realized by a class-C amplifier; **Only standard AM!!!**
- 2) **Low level modulation** - front stage small power AM wave;  
Achieved by analog multiplier; **(AM, DSB or SSB)**



# Diode Single Balance AM Circuit



$$u_{D1} = u_1 + u_2$$

$$u_{D2} = -u_1 + u_2$$



$$i_{D1} = g_D K(\omega_2 t) u_{D1}$$

$$i_{D2} = g_D K(\omega_2 t) u_{D2}$$

$$i_L = i_{D1} - i_{D2}$$

$$i_L = g_D K(\omega_2 t) u_{D1} - g_D K(\omega_2 t) u_{D2} = 2g_D K(\omega_2 t) u_1$$

# Frequency components of $i_L$



$$\begin{aligned} i_L &= g_D K(\omega_2 t) u_{D1} - g_D K(\omega_2 t) u_{D2} = 2g_D K(\omega_2 t) u_1 \\ &= 2g_D \left( \frac{1}{2} + \frac{2}{\pi} \cos \omega_2 t - \frac{2}{3\pi} \cos 3\omega_2 t + \frac{2}{5\pi} \cos 5\omega_2 t - \dots \right) U_1 \cos \omega_1 t \\ &= g_D U_1 \cos \omega_1 t + \frac{2}{\pi} g_D U_1 \cos(\omega_2 - \omega_1) t + \frac{2}{\pi} g_D U_1 \cos(\omega_2 + \omega_1) t \\ &\quad - \frac{2}{3\pi} g_D U_1 \cos(3\omega_2 - \omega_1) t - \frac{2}{3\pi} g_D U_1 \cos(3\omega_2 + \omega_1) t + \dots \end{aligned}$$

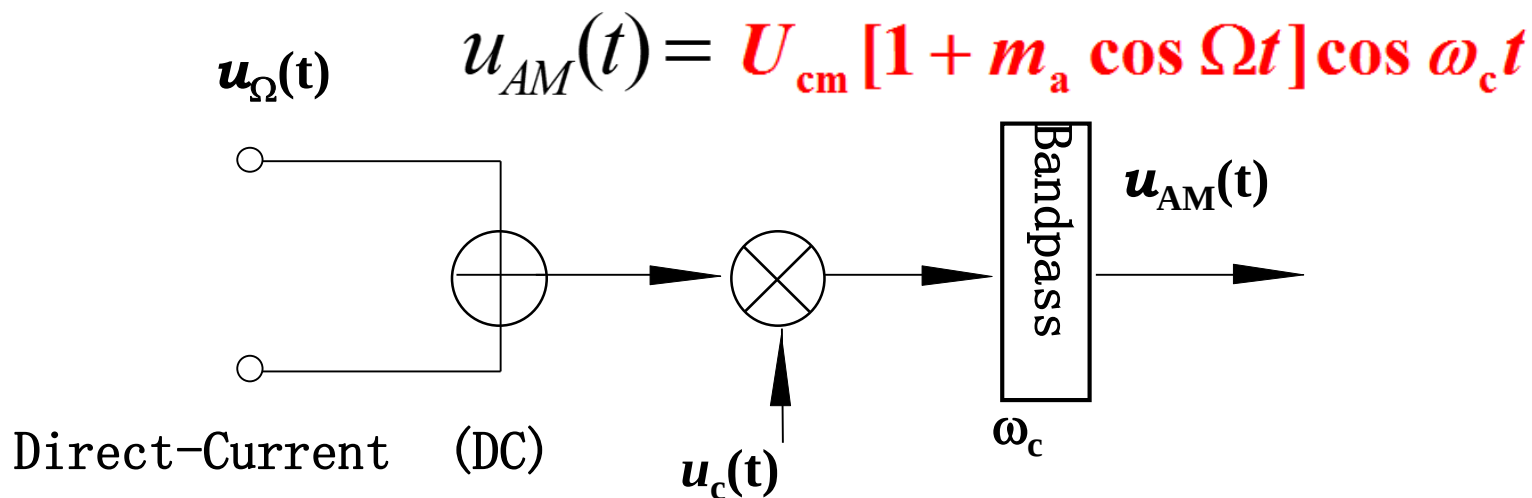
$$\omega_1, (\omega_2 \pm \omega_1), 3\omega_2 \pm \omega_1, \dots, (2n-1)\omega_2 \pm \omega_1$$
$$n = 1, 2, 3, \dots$$

**Q1: Whether diode balance circuit generate AM, DSB, SSB?**

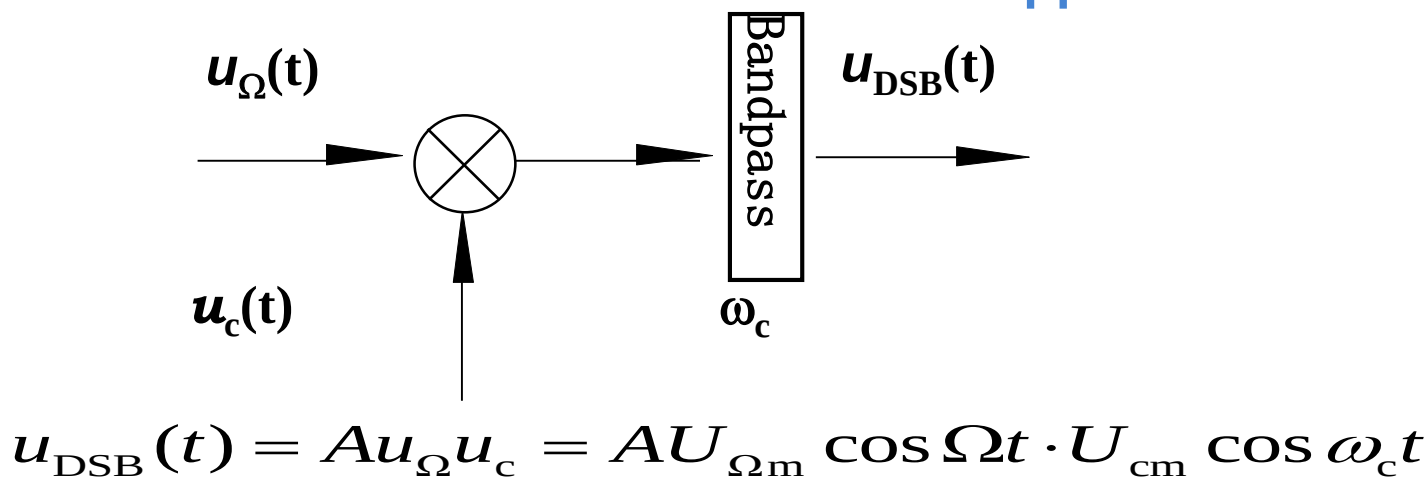
**Q2: How to generate?**

## 4.3.2 Low-level amplitude-modulated circuits

### (1) The realization model of standard AM wave



### (2) The realization model of DSB AM wave with suppressed carrier

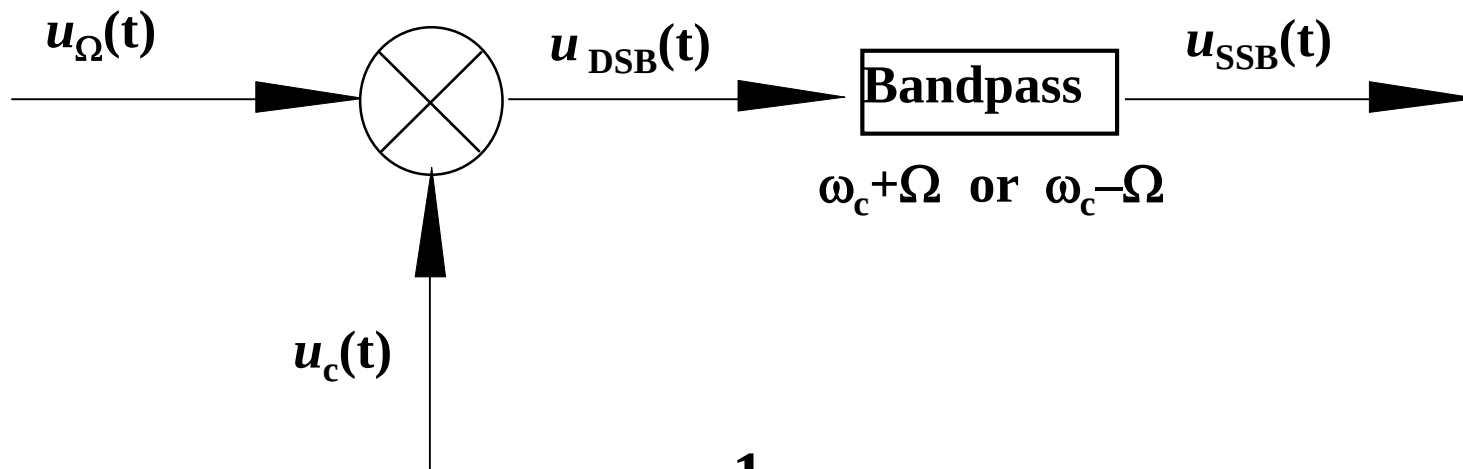






## 4.3.2 Low-level amplitude-modulated circuits

### (3) Single-sideband AM (SSB) model



$$u_{SSBH} = \frac{1}{2} A U_{\Omega m} U_{cm} \cos(\omega_c + \Omega)t$$

$$u_{SSBL} = \frac{1}{2} A U_{\Omega m} U_{cm} \cos(\omega_c - \Omega)t$$

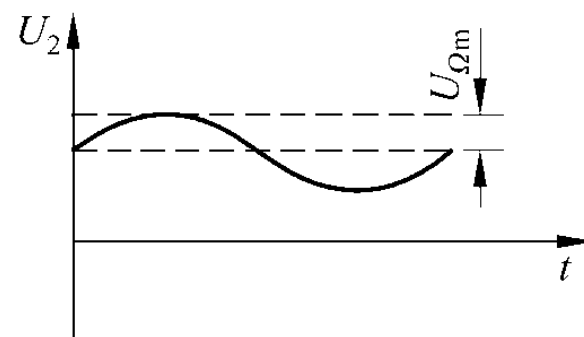
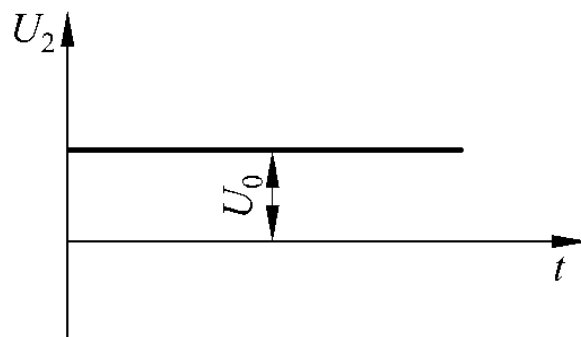
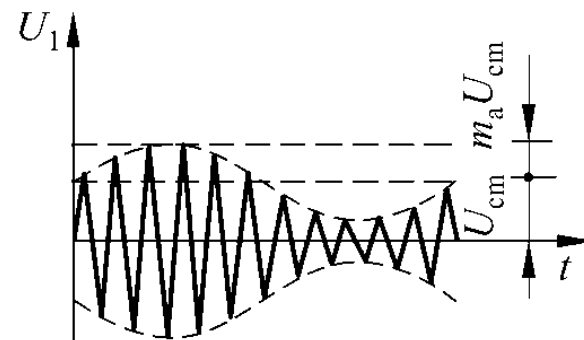
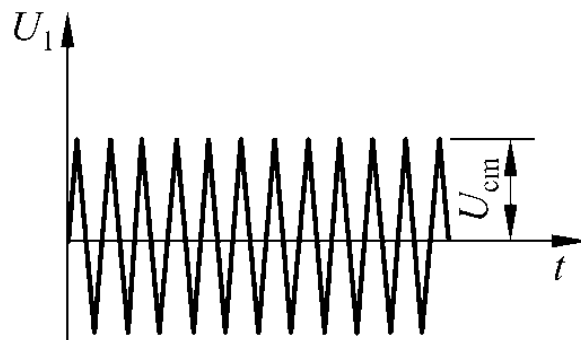
# 4 Demodulation circuit of AM wave



## ➤ Detection efficiency (Voltage transmission coefficient)

$$\eta_d = \frac{U_0}{U_{cm}}$$

$$\eta_d = \frac{U_{\Omega m}}{m_a U_{cm}}$$



## ➤ Detection distortion

$$\frac{5 \sim 10}{\omega_c} \leq R_L C \leq \frac{\sqrt{1 - m_{a\max}^2}}{m_{a\max} \Omega_{\max}}$$

$$2R_{in} \leq R_L \leq \frac{1 - m_{a\max}}{m_{a\max}} R_i$$

## ➤ Input resistance

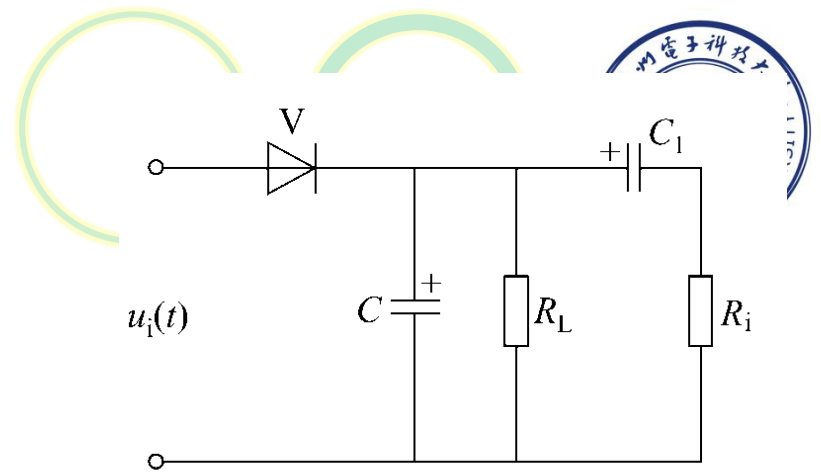
$$R_{in} = \frac{U_{cm}}{I_{1m}} \approx \frac{R_L}{2\eta_d} (\because \theta \approx 0, \therefore \frac{\alpha_1(\theta)}{\alpha_0(\theta)} \approx 2)$$

## ◆ Diagonal distortion

$$u_c(t) = U_{cm} [1 + m_a \cos \Omega t] e^{-\frac{t-t_A}{R_L C}}$$

$$\Rightarrow R_L C \leq \frac{\sqrt{1 - m_{a \max}^2}}{m_{a \max} \Omega_{\max}}$$

**No distortion condition**

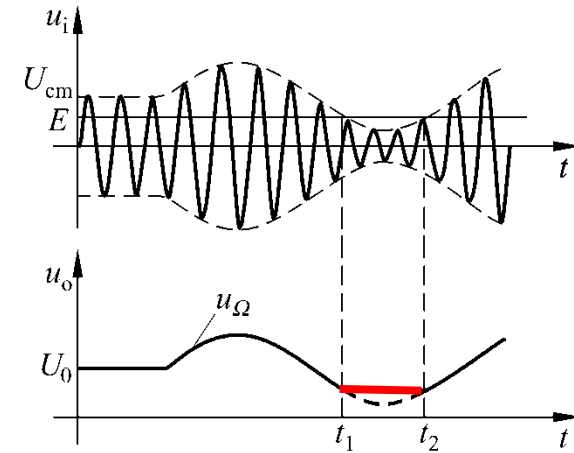
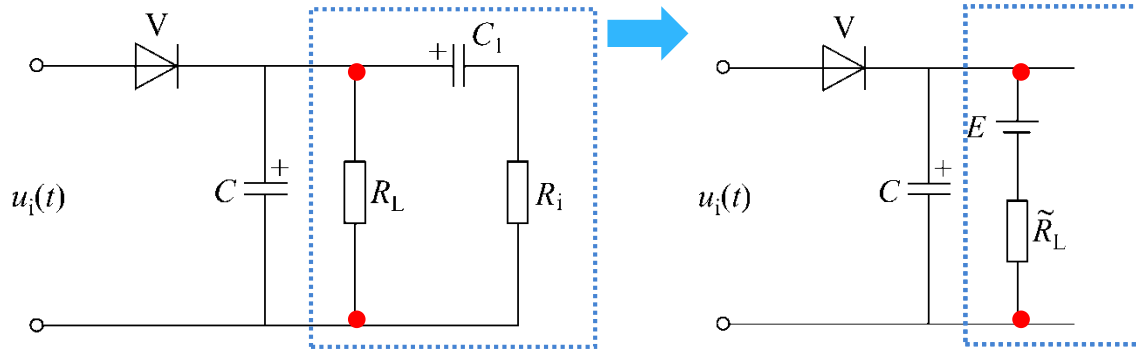
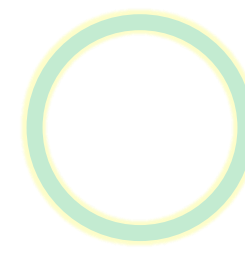
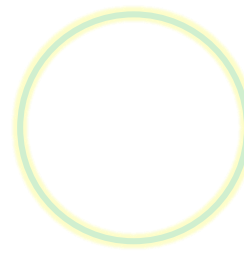


$CR_L \uparrow$  - Discharge slow

$m_a \uparrow$  - Modulation depth, envelope falling fast

$\Omega \uparrow$  - The period is short and the envelope is falling fast.

# ◆ Clipping distortion



$$E = \frac{R_L}{R_L + R_i} U_0 \quad \eta_d = \frac{U_0}{U_{cm}} \quad \tilde{R}_L = R_L // R_i$$

$$u_{AM}(t) = U_{cm} [1 + m_a \cos \Omega t] \cos \omega_c t$$

$$(1 - m_a) U_{cm} \geq E = \frac{R_L}{R_L + R_i} \eta_d U_{cm} \Rightarrow m_a \leq 1 - \eta_d \frac{R_L}{R_L + R_i}$$

$$\text{If } \eta_d = 1, \quad m_a \leq 1 - \frac{R_L}{R_L + R_i} = \frac{R_i}{R_L + R_i} = \frac{\tilde{R}_L}{R_L}$$

## 5) Selection and principle of detector



1. Selection of detector diode

2. Selection of  $R_L$ ,  $R_{in}$  and  $C$

$$\frac{5 \sim 10}{\omega_c} \leq R_L C \leq \frac{\sqrt{1 - m_{a \max}^2}}{m_{a \max} \Omega_{\max}}$$

$$2R_{in} \leq R_L \leq \frac{1 - m_{a \max}}{m_{a \max}} R_i$$

## 4.12 Interferences

### ➤ Whistle interference

$$f_{p,q} = |\pm p f_L \pm q f_c| \quad f_I = f_L - f_c$$

$f_{p,q} \approx f_I \pm F$   $F$  is a voice frequency (30-3400 Hz).

### ➤ Side-channel interference

$f_M$  is a frequency of external interference signal.

$$f_{p,q} = |\pm p f_L \pm q f_M| \quad f_{p,q} \approx f_I$$

$f_c$ : center frequency

$f_I$ : local oscillator freq

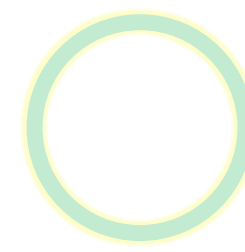
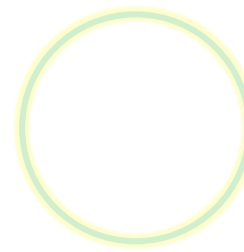
$f_M$ : interference freq

### ➤ Cross-modulation interference

The amplitude (envelope) of IF current is related to the amplitude of interference signal. ( $f_L$ ,  $f_c$ ,  $f_M$ )

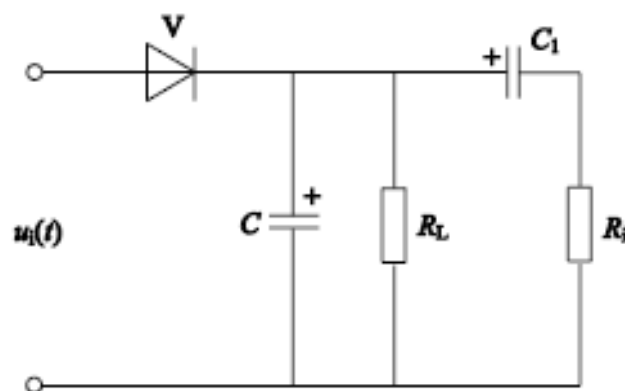
### ➤ Intermodulation interference

$$f_{p,q,r,s} = |\pm p f_L \pm q f_c \pm r f_{M1} \pm s f_{M2}| \quad f_{p,q,r,s} \approx f_I$$



The demodulation circuit is show in Ex.Fig.4.2 below with

$$u_i(t)=5\cos 2\pi\times 465\times 10^3t+4\cos 2\pi\times 10^3t \cos 2\pi\times 465\times 10^3t$$



Ex.Fig.4.2

The internal resistance of diode  $r_D=100\ \Omega$ ,  $C=0.01\mu\text{F}$  and  $C_1=47\mu\text{F}$ . In order to ensure no waveform distortion of output signal, please find

- (1) The maximum load resistance  $R_L$  of demodulator;
- (2) The minimum input resistance  $R_i$ .

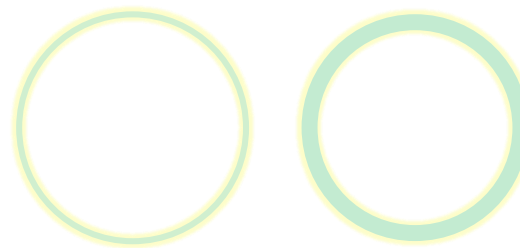
(1) according to the non-distortion condition ,we get

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} = \frac{0.6}{0.8} \times \frac{1}{2\pi \times 10^3 \times 0.01} = 11.9(\text{k}\Omega)$$

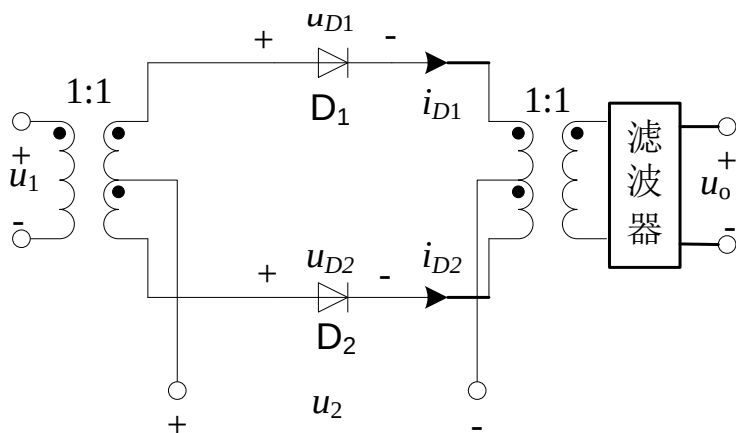
(2) according to the non-clipping distortion condition, we have

$$m_a \leq \frac{R_i}{R_i + R_L}$$

Put  $R_L=2.54\text{k}\Omega$  and  $m_a=0.8$  into formula above, we get  $R_i \geq 47.6\text{k}\Omega$



假设：二极管为理想二极管，导通电导为  $g_D$ ，滤波器阻抗为  $R_A$ ，请写出  $u_0(t)$  的表达式，画出其波形图和频谱图，并解释该电路的作用。



$$u_1(t) = U_{1m} (1 + m_a \cos 2\pi f_b t) \cos 2\pi f_1 t$$

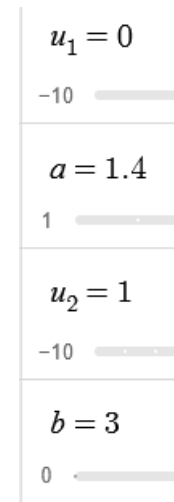
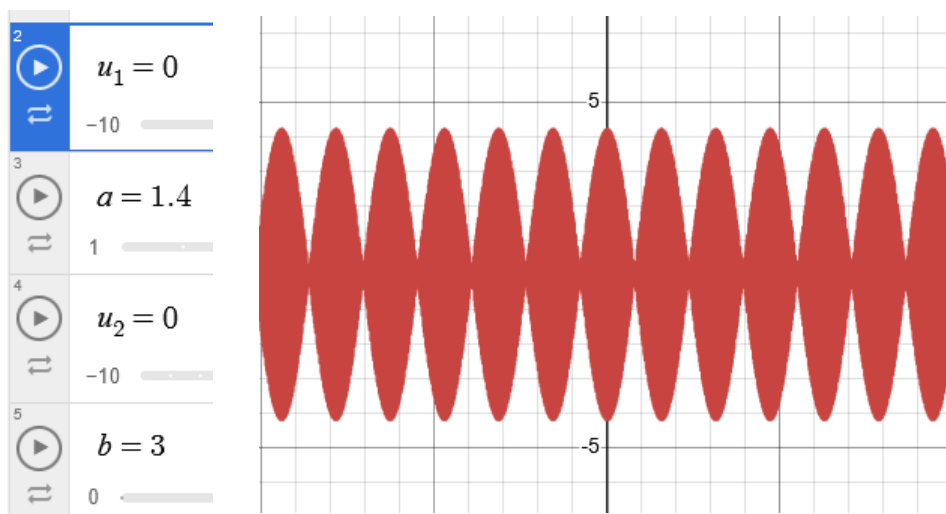
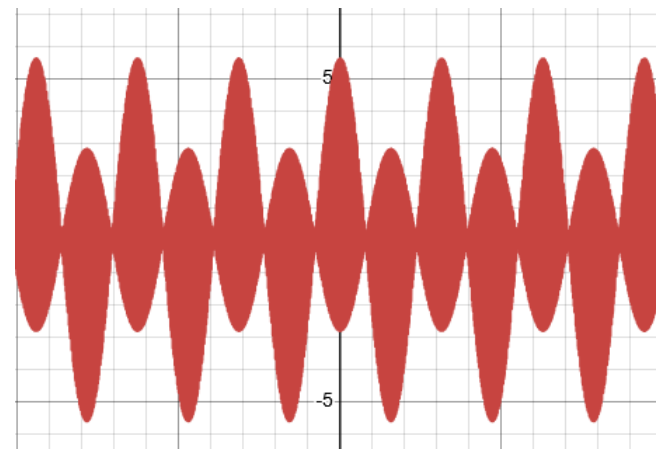
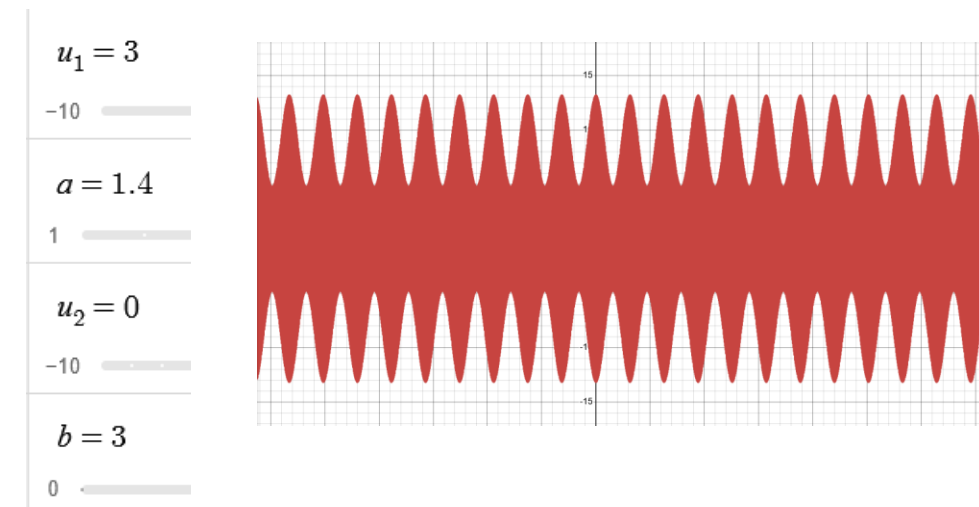
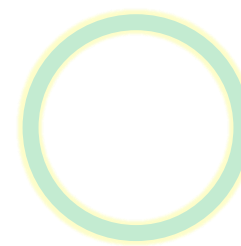
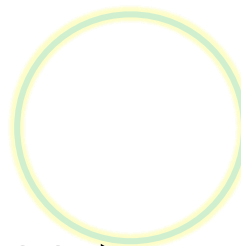
$$u_2(t) = U_{2m} \cos 2\pi f_2 t$$

$$U_{2m} \gg U_{1m}, f_1 \gg f_2 \gg f_b$$



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$$u_o(t) = (u_1 + a \cos 2t)(u_2 + b \cos 2000t)$$



# **Chapter 5 Angle modulation and demodulation**

**5.2 Properties of the angle modulation wave**

**5.3 Generation of angle modulation wave**

**5.4 FM circuits**

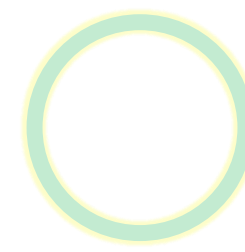
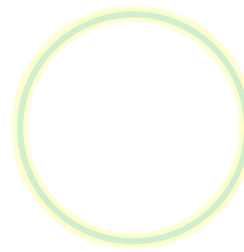
**5.7 Comparison of different modulation methods**

# Comparison between FM and PM (monofrequency modulation)



|                         | FM                                                                    | PM                                                     |
|-------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|
| Instantaneous frequency | $\omega(t) = \omega_c + K_f u_\Omega(t)$                              | $\omega(t) = \omega_c + K_p \frac{du_\Omega(t)}{dt}$   |
| Instantaneous phase     | $\theta(t) = \omega_c t + K_f \int u_\Omega(t) dt$                    | $\theta(t) = \omega_c t + K_p u_\Omega(t)$             |
| Maximum frequency shift | $\Delta\omega = K_f U_{\Omega m}$                                     | $\Delta\omega = K_p U_{\Omega m} \Omega$               |
| Maximum phase shift     | $m_f = \frac{K_f U_{\Omega m}}{\Omega} = \frac{\Delta\omega}{\Omega}$ | $m_p = K_p U_{\Omega m} = \frac{\Delta\omega}{\Omega}$ |
| Mathematical expression | $U_m \cos[\omega_c t + m_f \sin \Omega t]$                            | $U_m \cos[\omega_c t + m_p \cos \Omega t]$             |

# 1. Bandwidth of FM



The effective bandwidth of the FM is:

$$B = 2(m_f + 1)F = 2\Delta f_m + 2F$$

$$\Delta\omega = K_f U_{\Omega m}$$

$$m_f = \Delta\theta_{\max} = \frac{K_f U_{\Omega m}}{\Omega} = \frac{\Delta\omega_{\max}}{\Omega}$$

$$\left\{ \begin{array}{ll} K_f & \text{Rad/(s}\cdot\text{V)} \\ & \text{Hz/V} \\ U_{\Omega m} & \\ \Omega = 2\pi F & \end{array} \right.$$

$$(\Delta f_m = \frac{K_f U_{\Omega m}}{2\pi})$$

when  $m_f \ll 1$ , it is Narrow band frequency modulation(NBFM)  
bandwidth:  $B = 2F$

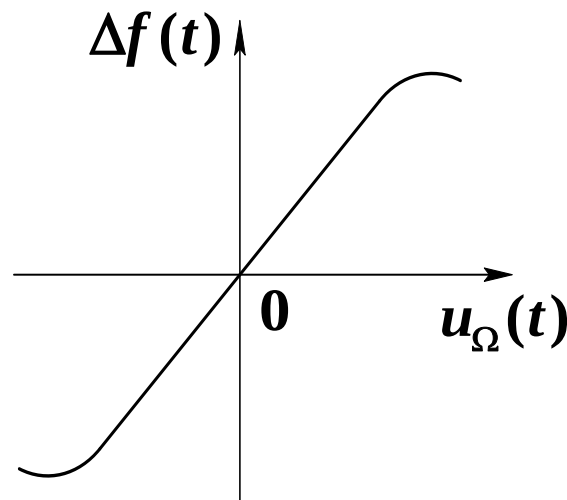


## 2 Modulation characteristics curve

### Modulation sensitivity

The unit modulation voltage caused frequency shift is modulation sensitivity.  $2\pi \cdot S = K_f$  in the linear section.

$$S = \left. \frac{d\Delta f(t)}{du_{\Omega}(t)} \right|_{u_{\Omega}(t)=0}$$



# 3. Comparison of modulation mode



## 1. Anti-interference ability (noise immunity)

FM is better than AM

## 2. Bandwidth occupation

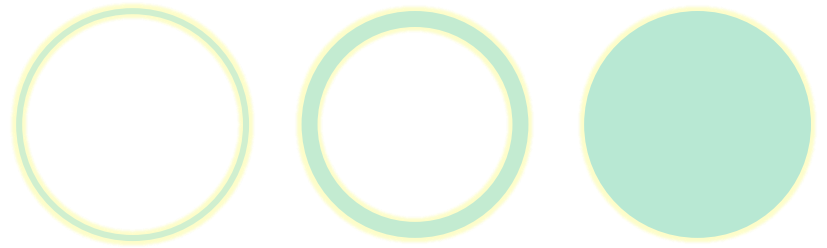
FM occupies **wider bandwidth** than AM. But it has better communication quality.

## 3. Transmitter power requirement

FM has **lower power consumption** than AM

## 4. Strong signal blockage

FM could avoid this blockage caused by strong signal to some extent.



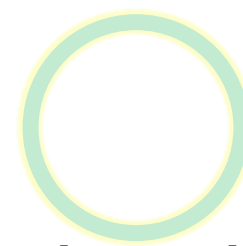
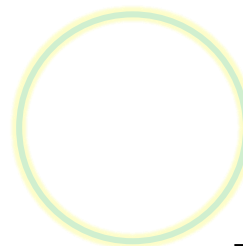
**Exercise1: For frequency modulation,  $K_f = 200$  Hz/V, if the modulation signal is shown below, analysis the effective bandwidth of the FM signal :**

**(a)  $U_{\Omega}(t) = 5\cos(2\pi \times 10^3 t) V$**

**(b)  $U_{\Omega}(t) = 10\cos(2\pi \times 10^3 t) V$**

**(c)  $U_{\Omega}(t) = 10\cos(4\pi \times 10^3 t) V$**

**(d)  $U_{\Omega}(t) = 5\cos(4\pi \times 10^3 t)V$**



**Exercise 2. For a carrier wave,  $u_c(t)=10 \cos 2\pi \times 10^5 t$ . The modulation signal is  $u_\Omega(t)=\cos 2\pi \times 10^3 t + 2\cos 2\pi \times 500 t$ . Please write out the expression of the FM wave, assuming that both modulation signals have the same  $\Delta f_{\max}=20$  KHz.**

$$u_{FM}(t) = U_{cm} \times \cos(2\pi \times f_c t + m_{f1} \sin \Omega_1 t + m_{f2} \sin \Omega_2 t)$$

$$\Delta f_{\max 1} = \Delta f_{\max 2} = \Delta f_{\max} = m_f \Omega / 2\pi = 20 \text{ KHz}$$

$$m_{f1} = (2\pi \times \Delta f_{\max}) / \Omega_1 = 2\pi \times 20 \times 10^3 / 2\pi \times 10^3 = 20,$$

$$m_{f2} = (2\pi \times \Delta f_{\max}) / \Omega_2 = 2\pi \times 20 \times 10^3 / 2\pi \times 500 = 40$$

$$u_{FM}(t) = U_{cm} \times \cos(2\pi \times 10^8 t + m_{f1} \sin \Omega_1 t + m_{f2} \sin \Omega_2 t)$$

$$= 5 \cos(2\pi \times 10^8 t + 20 \sin 2\pi \times 10^3 t + 40 \sin 2\pi \times 500 t)$$